Meta Quest 3 UVC Camera / Overlay Keyboard Issue — Investigation Summary

Platform: Meta Quest 3 / Horizon OS
Application: Unity application using OpenXR and a UVC/USB camera plugin
Relevant plugin: `com.serenegiant.unity.uvcplugin`
Date investigated: August 13–14, 2026

Expected behavior

The application should:

1. Detect/use an externally connected UVC USB camera.
2. Request whatever permissions are necessary to access it.
3. Allow the user to launch the application normally.
4. Ideally **not** present an "Open with..." / application chooser every time a camera is connected.
5. Optionally use the Quest overlay keyboard for text input.

Relevant Android manifest

The application currently has:

```
<uses-feature
    android:glEsVersion="0x00020000"
    android:required="true"/>

<uses-feature
    android:name="android.hardware.usb.host"
    android:required="true"/>

<uses-feature
    android:name="android.hardware.camera"
    android:required="false"/>

<uses-feature
    android:name="oculus.software.overlay_keyboard"
    android:required="false"/>

<uses-permission
    android:name="horizonos.permission.USB_CAMERA"/>
```

The UVC plugin also declares:

```
<activity
    android:name="com.serenegiant.unity.uvcplugin.UsbPermissionActivity"
    android:theme="@android:style/Theme.NoDisplay"
    android:exported="true">

    <intent-filter>
        <action
            android:name="android.hardware.usb.action.USB_DEVICE_ATTACHED"/>
    </intent-filter>

    <meta-data
        android:name="android.hardware.usb.action.USB_DEVICE_ATTACHED"
```

```
        android:resource="@xml/device_filter_uvc"/>

</activity>
```

The `device_filter_uvc.xml` resource determines which USB devices match the UVC attachment handler.

Initial problem: intrusive USB app chooser

When a UVC camera is connected, Quest/Android can show a chooser asking which application should handle the video device.

This appears to be related to:

```
<action android:name="android.hardware.usb.action.USB_DEVICE_ATTACHED"/>
```

combined with the USB device filter.

Multiple installed applications supporting UVC cameras appeared in the chooser.

I subsequently removed other applications that registered similar camera/USB handling capabilities. **After doing this, camera functionality unexpectedly became easier to recover and/or more reliable.**

This suggests that the Quest USB device association/permission state may be involved.

Overlay keyboard investigation

I added:

```
<uses-feature
    android:name="oculus.software.overlay_keyboard"
    android:required="false"/>
```

because the application reported:

> `oculus overlay keyboard is disabled, add oculus.software.overlay_keyboard feature request to your android manifest`

After adding this feature, the **USB camera permission/access behavior appeared to break**.

The camera could no longer reliably be accessed.

Removing the feature did not immediately restore camera functionality. Recovery appeared inconsistent and sometimes required reinstalling/removing applications and/or resetting the environment.

This initially suggested a possible interaction between:

```
oculus.software.overlay_keyboard
```

and:

```
UVC / USB camera permission handling
```

However, this has **not been proven to be the direct cause**.

OpenXR investigation

I discovered an unintended OpenXR feature in the Unity project:

```
m_Name: SessionStateFeature Standalone
m_Enabled: 0
nameUi: Session State Tracker
version: 1.0.0
featureIdInternal: com.beanotherlab.presence
openxrExtensionStrings: XR_test
company: BeAnotherLab
priority: 0
required: 0
```

This appeared suspicious because it was not intentionally added.

However, the feature has:

```
m_Enabled: 0
```

and **the camera problem persists when this OpenXR feature is disabled**.

Therefore, at present there is **no evidence that this OpenXR feature is responsible**.

It should not currently be considered the primary suspect.

Horizon OS update

A Quest update was installed on **August 12, 2026**, immediately before/around the period in which these issues were being investigated.

The public Meta release information checked so far does **not explicitly mention a UVC-camera, USB-permission, or overlay-keyboard change** corresponding to the update.

Therefore, a Horizon OS regression/change is **possible but unconfirmed**.

The exact Quest OS/build number should be recorded for a Meta support report.

Current observations

The most important observations are:

1. Camera behavior is inconsistent

The same application/configuration can sometimes work and sometimes fail after seemingly minor changes.

2. Removing other UVC-capable applications helped

Previously installed applications that could handle similar USB/video devices were removed.

Afterward, previously lost camera functionality could be recovered.

This suggests that **USB device association, handler selection, or permission state may be involved**.

3. The overlay keyboard feature correlates with camera failures

Adding:

```
<uses-feature
    android:name="oculus.software.overlay_keyboard"
    android:required="false"/>
```

appeared to interfere with the camera permission/access flow.

However, because removing it did not immediately restore the camera, this may be a **stateful/system interaction rather than a direct incompatibility**.

4. OpenXR feature does not appear to be the cause

The suspicious `SessionStateFeature Standalone` is disabled, and the problem persists.

5. The USB attachment handler is a significant part of the architecture

The application registers:

```
android.hardware.usb.action.USB_DEVICE_ATTACHED
```

and provides:

```
@xml/device_filter_uvc
```

This is likely responsible for the automatic "which app should open this camera?" behavior.

Current hypotheses

In approximate order of interest:

1. Quest/Horizon OS USB device association or permission state

The strongest candidate given the inconsistent behavior and the improvement after removing competing UVC applications.

2. Interaction between UVC permission handling and Quest's overlay keyboard feature

The correlation is strong enough to investigate, but there is currently no explanation for why these unrelated features would interact.

3. Horizon OS update/regression

The August 12 update makes this worth investigating, particularly if the exact OS build changed USB/camera behavior.

4. Unity/OpenXR manifest/build configuration

Possible, but the suspicious OpenXR feature appears disabled and hasn't explained the behavior.

5. UVC plugin behavior

`UsbPermissionActivity` and its USB attachment/permission flow may be interacting with Quest's newer Android/Horizon OS behavior.

Useful next diagnostic steps

The most useful comparison would be between a **known-good APK** and a **known-bad APK**.

Inspect their actual packaged manifests:

```
apkanalyzer manifest print working.apk > working-manifest.txt
apkanalyzer manifest print broken.apk > broken-manifest.txt
```

Then:

```
diff -u working-manifest.txt broken-manifest.txt
```

Pay particular attention to:

```
oculus.software.overlay_keyboard
android.permission.CAMERA
horizonos.permission.USB_CAMERA
android.hardware.usb.action.USB_DEVICE_ATTACHED
UsbPermissionActivity
```

On the Quest, capture:

```
adb shell dumpsys usb
```

and:

```
adb shell dumpsys package YOUR.PACKAGE.NAME
```

while the camera is connected and the problem is occurring.

Also record:

```
adb shell getprop ro.build.version.release
adb shell getprop ro.build.version.incremental
```

to establish the exact Horizon OS/Android build.

Concise issue description for Meta

> **Quest 3 UVC camera permission/USB association becomes inconsistent, apparently correlated with `oculus.software.overlay_keyboard`**

We have a Unity/OpenXR application using a UVC USB camera plugin (`com.serenegiant.unity.uvcplugin`). The app registers `USB_DEVICE_ATTACHED` and uses a UVC device filter.

The Quest sometimes presents an application chooser when a USB video device is connected because multiple installed applications can handle the device. Removing other UVC-capable applications unexpectedly improved/recovered camera functionality.

Separately, adding the following manifest feature:

```
<uses-feature
    android:name="oculus.software.overlay_keyboard"
    android:required="false"/>
```

appears to interfere with the UVC camera permission/access flow. Removing the feature does not reliably restore camera access without additional state changes/reinstallation/device resets.

An unrelated OpenXR feature (`com.beanotherlab.presence`, `SessionStateFeature Standalone`) was also discovered in the Unity project, but it is disabled (`m_Enabled: 0`) and the issue persists when it is disabled.

A Horizon OS update was installed on August 12, 2026, shortly before the issue was investigated. Public release notes do not identify a specific UVC/USB-camera change.

We would like to know whether there are known Horizon OS issues involving:

- UVC camera USB permission persistence
- multiple applications registering `USB_DEVICE_ATTACHED`
- USB video device application association
- `horizonos.permission.USB_CAMERA`
- `oculus.software.overlay_keyboard`
- interaction between the Quest overlay keyboard and USB camera permissions
- changes/regressions in the August 2026 Horizon OS update

Current status: camera functionality can be recovered, but the conditions for recovery are inconsistent and not yet understood. The overlay keyboard feature is currently being treated as a potential trigger, not a confirmed root cause.